

12.7 Fractional indices

Navigator

Answers
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Q1 Column 1, Q2 Columns 1–2,
Q3, Q4, Q5 (a)–(d), Q7 (a)–(d),
Q9, Q10, Q11, Q12

Q1 Column 2, Q2 Columns 2–3,
Q3, Q4, Q5 Column 1, Q6, Q7
(c)–(f), Q9, Q10, Q11, Q12

Q1 Column 3, Q2 Columns 3–4,
Q3, Q5 Column 2, Q6, Q7 (e)–(h),
Q8, Q9, Q10, Q11, Q13

Equipment required: calculator for Questions 2, 3, 4, 7, 10, 11 and 12

Fluency

WE13

1 Simplify each of the following.

(a) $x^{\frac{1}{3}} \times x^{\frac{1}{3}}$

(b) $a^{\frac{1}{2}} \times a^{\frac{1}{3}}$

(c) $k^{\frac{1}{3}} \times k^{\frac{1}{5}}$

(d) $\frac{m^{\frac{1}{2}}}{m^{\frac{1}{4}}}$

(e) $\frac{p^{\frac{1}{2}}}{p^{\frac{1}{3}}}$

(f) $\left(y^{\frac{1}{2}}\right)^{\frac{1}{2}}$

(g) $(25p^4)^{\frac{1}{2}}$

(h) $(49x^6)^{\frac{1}{2}}$

(i) $(8ab^3)^{\frac{1}{3}}$

(j) $\sqrt{x^4y}$

(k) $\sqrt{kn^8}$

(l) $\left(\frac{p^4}{q^{10}}\right)^{\frac{1}{2}}$

WE14

2 Evaluate each of the following. Express your answer correct to two decimal places where appropriate.

(a) $196^{\frac{1}{2}}$

(b) $529^{\frac{1}{2}}$

(c) $6859^{\frac{1}{3}}$

(d) $512^{\frac{1}{3}}$

(e) $1296^{\frac{1}{4}}$

(f) $4096^{\frac{1}{4}}$

(g) $3125^{\frac{1}{5}}$

(h) $1024^{\frac{1}{5}}$

(i) $500^{\frac{1}{3}}$

(j) $3000^{\frac{1}{3}}$

(k) $98^{\frac{1}{6}}$

(l) $145^{\frac{1}{6}}$

3 Using your calculator to assist you, place these numbers in ascending order.

$2^{\frac{1}{2}}, 3^{\frac{1}{3}}, 4^{-1}, 6^{-\frac{1}{5}}$

4 (a) 3^5 is closest to:

A 1.71

B 15

C 125

D 243

(b) 4^{-3} is closest to:

A 0.012

B 0.016

C 64

D 81

(c) $20^{\frac{1}{4}}$ is closest to:

A 0.000 006

B 2.1

C 5

D 160 000

(d) $75^{\frac{1}{3}}$ is closest to:

A -25

B 0.24

C 4.22

D 421 875

- 5 Simplify each of the following expressing your answers with positive indices.

(a) $\frac{\left(a^{\frac{1}{3}}\right)^6 \times a}{a^2}$

(b) $\frac{\left(d^{\frac{1}{2}}\right)^8 \times d^2}{d^3}$

(c) $(x^4 y^6)^{\frac{1}{2}} \times (x^{-3} y^{-12})^{\frac{1}{3}}$

(d) $(m^6 n^3)^{\frac{1}{3}} \times (m^{-8} n^{-10})^{\frac{1}{2}}$

(e) $\frac{(pq^3)^{\frac{1}{2}} \times p^{\frac{1}{2}} q^{\frac{1}{2}}}{pq^4}$

(f) $\frac{(a^5 b)^{\frac{1}{2}} \times a^{\frac{1}{2}} b^{\frac{1}{2}}}{a^3 b^5}$

(g) $\left(\frac{m^2 n}{n^{\frac{1}{2}}}\right)^{-\frac{1}{3}} \times \frac{m\left(n^{\frac{1}{3}}\right)^2}{2n^{\frac{1}{3}}}$

(h) $\left(\frac{x^3 y^2}{y^{\frac{1}{2}}}\right)^{-\frac{1}{3}} \times \frac{(x^2)^{\frac{1}{3}} y}{5y^{\frac{2}{3}}}$

- 6 (a) Find the expression that is *not* the same as $a^{\frac{3}{2}} b^{\frac{1}{2}}$.

A $\sqrt{a^3 b}$

B $(\sqrt{ab})^3$

C $(a^3 b)^{\frac{1}{2}}$

D $(\sqrt{a})^3 \sqrt{b}$

- (b) Simplifying $\left(\frac{a^{\frac{1}{2}} b^3}{b^2}\right)^2 \div \frac{b^{\frac{3}{2}} a}{b^4}$ gives:

A $b^{\frac{9}{2}}$

B $b^{\frac{13}{2}}$

C $\frac{b^{\frac{7}{2}}}{a^{\frac{1}{2}}}$

D $\frac{b^7}{a^{\frac{1}{2}}}$

- 7 Evaluate each of the following correct to three decimal places.

(a) $8^{-\frac{1}{3}}$

(b) $12^{-\frac{1}{3}}$

(c) $37^{-\frac{1}{2}}$

(d) $26^{-\frac{1}{2}}$

(e) $54^{-\frac{1}{4}}$

(f) $72^{-\frac{1}{4}}$

(g) $100^{-\frac{1}{5}}$

(h) $167^{-\frac{1}{5}}$

Understanding

- 8 By simplifying first, find the value of x in each of the following equations.

(a) $\frac{72x^2}{\left(x \times x^{\frac{3}{2}}\right)^2} = 24$

(b) $\frac{\left(x^{\frac{1}{2}}\right)^5}{x^{-2}} \times \frac{5x^{-1}}{x^2} = 40$

(c) $\frac{7x^{-3}}{x^3} \left(\frac{x^{\frac{3}{2}}}{x}\right)^6 = 56$

- 9 Simplify the following.

(a) $\sqrt{x^6}$

(b) $\sqrt[3]{a^9}$

(c) $\sqrt[4]{16m^8n^4}$

Reasoning

- 10 Find the value of x in each of the following equations.

(a) $x^{-2} = \frac{1}{4}$

(b) $3^x = 27$

(c) $3^x = \frac{1}{9}$

(d) $2 \times 5^x = 50$